

Define: good/bad pose ,video

Aim:

- With the use of action recognition in home or office to either keep track of our behaviors (sitting/ falling) so as to improve our health, enhance the safety and work promptly.
- NO additional large device (such as computer), but a small pi and camera are required
- NO lots of sensors set-ups, just a camera

Procedures:

- 1) Raspberry **Pi camera to get the images** (fps)
- 2) Use **OpenPose/** to extract features (**RGB to Skeleton data**)
 - OpenPose pre-trained model, the algorithm will extract 18 body joints and 17 lines connecting joints.
 - +ve 1: focus on correct foreground (no need depth analysis and extraction)
 - +ve 2: could higher accuracy than directly train?

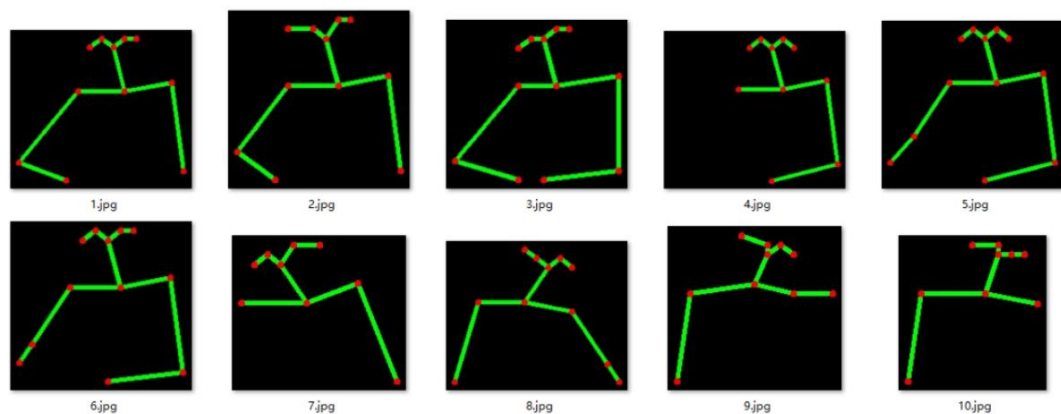


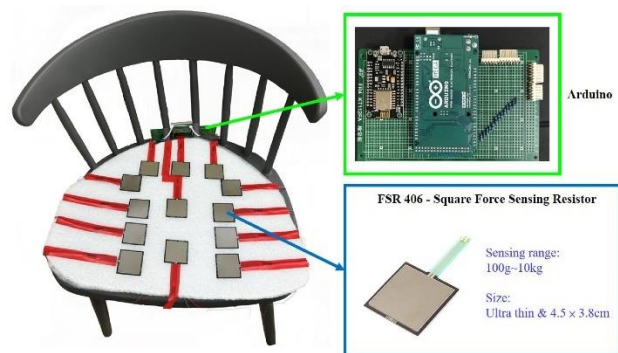
Figure 2. Dataset of Incorrect Sitting Posture.

<https://iopscience.iop.org/article/10.1088/1757-899X/677/3/032057/pdf>

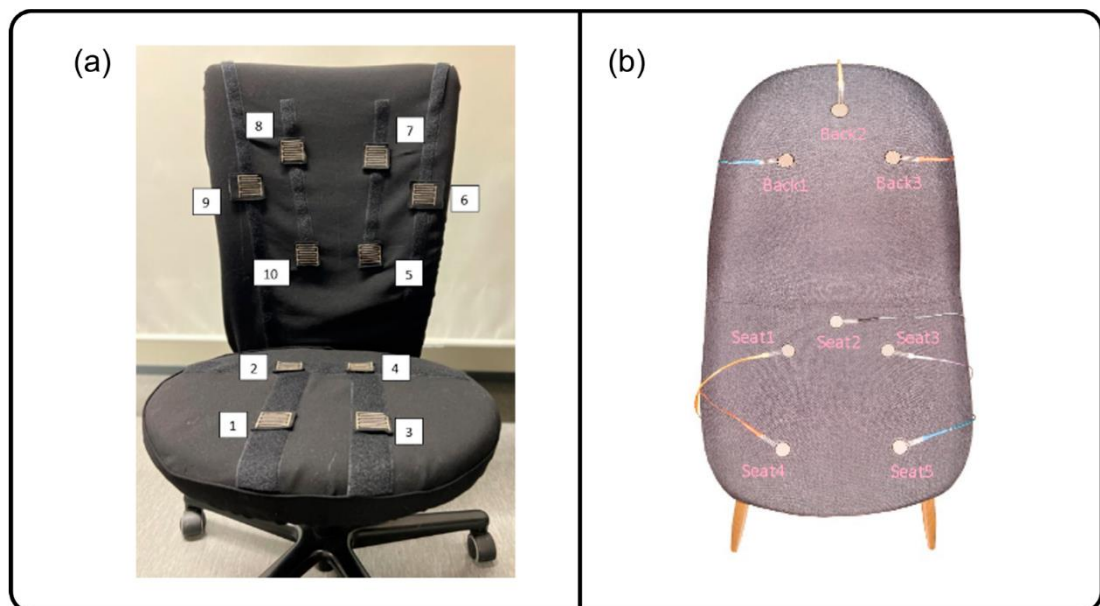
3) Pass to CNN/ ViT to perform classifications

⇒ All Deploy in pi?

other sources --- Previous work



1) (pressure sensor) :



Claim high accuracy, but we need to set the chair one by one like this – lots of sensors (13 sensors)?

<https://www.mdpi.com/1424-8220/23/13/5894>

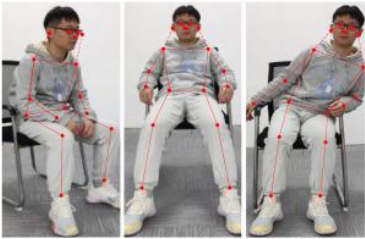
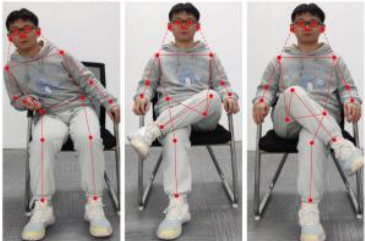
<https://www.mdpi.com/1424-8220/24/9/2940>

force sensor, textile pressure sensor,

2) OpenCV

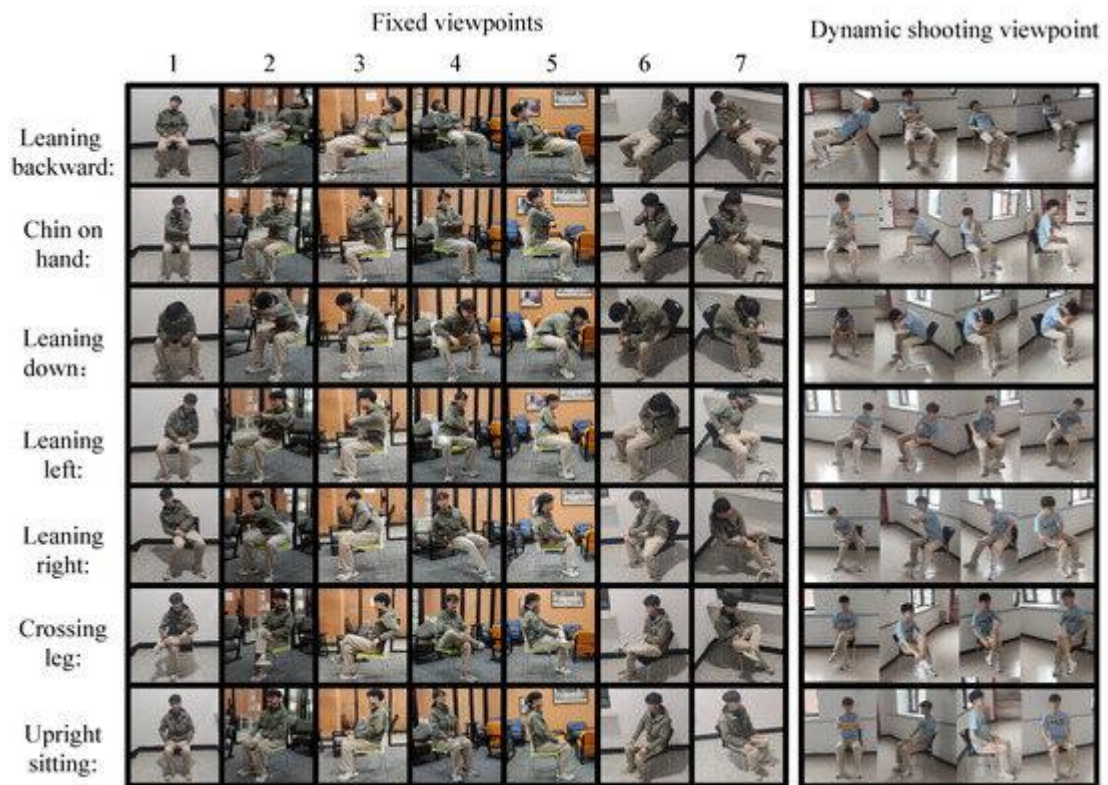
<https://learnopencv.com/wp-content/uploads/2022/03/MediaPipe-pose-feature-image.gif>

-ve: only can show good pose or bad pose (limited labels?)

Label	Example image that represents the label	
Buttocks_slipping		
Backbone_curved		
Neck_bending		



https://www.researchgate.net/publication/345842487_3D_PostureNet_A_unified_framework_for_skeleton-based_posture_recognition



<https://www.mdpi.com/2079-9292/14/4/718#B22-electronics-14-00718>

neck and torso inclination



Neck Inclination : 29
Torso Inclination : 34

RFID approach

https://www.researchgate.net/publication/353087625_Sitsen_Passive_sitting_posture_sensing_based_on_wireless_devices

Sitsen, a non-contact wireless-based sitting posture recognition system, just using radio frequency signals alone, which neither compromises the privacy nor requires using various specific sensors.



Figure 2. The five habitual sitting postures. (a) Posture A means sitting straight. (b) Posture B means back bend forward. (c) Posture C means sitting backward. (d) Posture D means the waist is straight forward and the waist and thigh are at an angle of 30°. (e) Posture E means the waist is straight forward and the waist and thigh are at an angle of 45°.